

CS & IT ENGINEERING



THEORY OF COMPUTATION

Regular Language

Lecture No. - 02



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Recap of Previous Lecture



Topic

Regular language Detection
?????

finite language $\Rightarrow$ Regular

Infinite language $\begin{matrix} \rightarrow \text{Regular} \\ \rightarrow \text{Non Regular} \end{matrix}$



Topics to be Covered



Topic

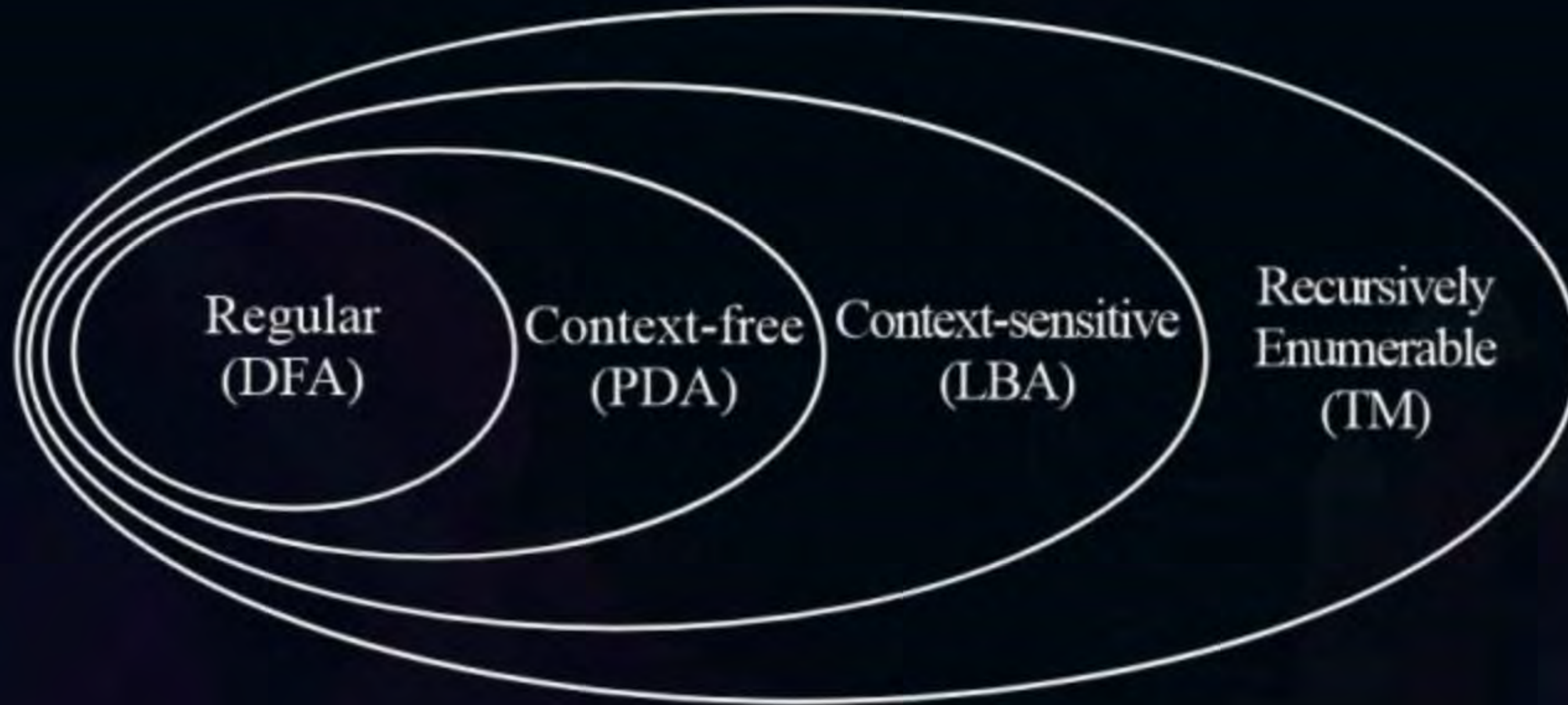
Regular Language Detection

Topic

Pumping Lemma



Topic : Theory of Computation





Topic : Regular Language Detection

Which of these languages are Regular?

$$L = \{a^n b^n c^n \mid 1 \leq n \leq 1000\}$$

$$L = \{a^n b^m \mid n + m = 10\}$$

$$L = \{a^n b^m \mid n - m = 5\}$$

$$L = \{a^n b^m \mid n \times m = 100\}$$

$$L = \{a^n b^m \mid n = 2m + 1\}$$

$$L = \{a^n b^m \mid n > m\}$$

$$L = \{a^n b^m \mid n > m \text{ and } n < m\}$$



Topic : Regular Language Detection

- Every finite language is regular.
- Every regular language need not be finite.
- All palindrome languages over more than one symbol are non regular.
- Any infinite language require infinite memory then it is non regular.



Topic : Regular Language Detection

$$L = \{a^n b^m \mid n > m \text{ (or) } n < m\}$$

$$L = \{a^n b^m c^{n+m} \mid n, m \geq 1\}$$

$$L = \{a^n b^{n+m} \mid n, m \geq 1\}$$

$$L = \{a^n b^{2m} c^{3k} \mid n, m, k \geq 0\}$$

$$L = \{a^n b^{m^2} c^{k^3} \mid n, m \geq 1\}$$

$$L = \{a^{2^n} \mid n \geq 0\}$$

$$L = \{a^{n^2} \mid n \geq 1\}$$



Topic : Regular Language Detection

$$L = \{a^{n^n} \mid n \geq 1\}$$

$$L = \{a^{100^{100^{100}}} \mid\}$$

$$L = \{(a^p)^* \mid p \text{ is prime number}\}$$

① $L = \{a^p \mid p \text{ is prime number}\} \rightarrow \text{Non Regular}$

② $L = \{a^k \mid k \text{ is even number}\} \rightarrow \text{Regular}$

$$L = \{ww^R \mid w \in \{a, b\}^*\}$$

$$L_1 = \{ W W^R \mid W \in (a+b)^* \} \Rightarrow \text{Non Regular}$$

$$L_2 = \{ W W^R \mid W \in (a)^* \} \Rightarrow \text{Regular}$$

$$L_3 = \{ \overrightarrow{W} c \overleftarrow{W}^R \mid W \in (a+b)^* \} \Rightarrow \text{Non Regular}$$

$$L_4 = \{ W x W^R \mid W, x \in (a+b)^* \} \Rightarrow \text{Regular}$$

(a+b)*

Regular

$$L_5 = \left\{ \underbrace{w}_{abba} \underbrace{w^R}_{(a+b)^*} x \mid w, x \in (a+b)^* \right\} = \underline{(a+b)^*}$$

$$L_6 = \left\{ \underbrace{x}_{(a+b)^*} \underbrace{w}_{\epsilon} \underbrace{w^R}_{\epsilon} \mid w, x \in (a+b)^* \right\} \Rightarrow (a+b)^*$$

Non Reg

$$L_7 = \left\{ \underbrace{w}_{abba} \underbrace{w^R}_{(a+b)^+} x \mid w, x \in (a+b)^+ \right\} \Rightarrow \underline{\underline{\text{Non Regular}}}$$

$(a+b)^+(aa+bb)$

$$L_8 = \left\{ \underbrace{x}_{(a+b)^+} \underbrace{w}_{abba} \underbrace{w^R}_{(a+b)^+} \mid w, x \in (a+b)^+ \right\} \Rightarrow \underline{\underline{\text{Non Reg}}}$$

$$L_9 = \{W X W^R \mid W, X \in (a+b)^*\} \Rightarrow \underline{\text{Regular}}$$

$$L_{10} = \left\{ \overbrace{W X W^R}^{\text{ababaaababaa}} \mid W, X \in (a+b)^+ \right\} \Rightarrow \underline{\text{Regular}}$$

W = min $a(a+b)^+a + b(a+b)^+b$

(Q) Which of the following is regular?

(a) $\{WW^RX\}$ N.R

(b) $\{XWW^R\}$ N.R

(c) $\{WXW^R\}$ regular

(d) $\{WW\}$ Non-R

$W, X \in (a+b)^+$

$$x \in (a+b)^*$$

$$L_1 = \{x \mid n_a(x) = n_b(x)\} \Rightarrow \text{Non Reg}$$

$$L_2 = \{x \mid n_a(x) > n_b(x)\} \Rightarrow \text{Non Reg}$$

$$L_3 = \{x \mid n_a(x) \neq n_b(x)\} \Rightarrow \text{Non Reg}$$

$$x, y \in (a+b)^*$$

$$L_4 = \{ xy \mid n_a(x) = n_b(y) \} \Rightarrow \text{Regular}$$

$(a+b)^*$

$$\{ \epsilon \epsilon, \epsilon a, b \epsilon, \epsilon \text{aa}, bb, ab, ba, \dots \}$$

$x \quad y \quad x \quad y$

$$L_5 = \{ x \$ y \mid n_a(x) = n_b(y) \} \Rightarrow \text{Non Regular}$$

$x, y \in (a+b)^*$



Topic : Regular Language Detection

$$L = \{wbw^R \mid w \in \{a\}^*\}$$

$$L = \left\{ x \mid x \in \{a, b\}^* \right. \\ \left. n_a(x) \bmod 3 = n_b(x) \bmod 3 \right\}$$

$$L = \left\{ x \mid x \in \{a, b\}^* \right. \\ \left. n_a(x) \bmod 3 > n_b(x) \bmod 3 \right\}$$

$$L = \left\{ x \mid x \in \{a, b\}^* \right. \\ \left. n_a(x) \bmod 3 \neq n_b(x) \bmod 3 \right\}$$

$$L = \left\{ x \mid x \in \{a, b, c\}^* \right. \\ \left. n_a(x) \neq n_b(x) \right\}$$



Topic : Regular Language Detection

- Set of all balanced parentheses
- Equal number of open and close parentheses
- Even length palindrome strings of English Language.
- Odd length palindrome strings of Hindi Language

#Q. Consider the following languages:

$$L_1 = \{ww \mid w \in \{a, b\}^*\} \rightarrow \text{N.R.}$$

$$L_2 = \{ww^R \mid w \in \{a, b\}^*, w^R \text{ is the reverse of } w\} \rightarrow \text{N.R.}$$

$$L_3 = \{0^{2i} \mid i \text{ is an integer}\} \rightarrow \text{Reg.}$$

$$L_4 = \{0^{i^2} \mid i \text{ is an integer}\} \rightarrow \text{N.R.}$$

Which of the language are regular ?

A Only L_1 and L_2

B Only L_2 , L_3 and L_4

C Only L_3 and L_4

D Only L_3

#Q. Which of the following are regular sets?

1. $\{a^n b^{2m} \mid n \geq 0, m \geq 0\} \rightarrow \text{Reg}$
2. $\{a^n b^m \mid n = 2m\} \rightarrow \text{N.R}$
3. $\{a^n b^m \mid n \neq m\} \rightarrow \text{N.R}$
4. $\{xcy \mid x, y \in \{a, b\}^*\} \rightarrow \text{regular}$

A 1 and 4 only

B 1 and 3 only

C 1 only

D 4 only

#Q. Which of the following languages is/are regular?

$\{wxw^R\}$ $\rightarrow$ regular

$L_1 : \{wxw^R \mid w, x \in \{a, b\}^+ \text{ and } |w|, |x| > 0 \text{ } w^R \text{ is the reverse of string } w\}$

$L_2 : \{a^n b^m \mid m \neq n \text{ and } m, n \geq 0\} \rightarrow \text{N.R.}$

$L_3 : \{a^p b^q c^r \mid p, q, r \geq 0\} \rightarrow \text{Reg}$

A L_1 and L_3 only

B L_2 only

C L_2 and L_3 only

D L_3 only

#Q. Which one of the following is TRUE?

- ☒ A The language $L = \{a^n b^n \mid n \geq 0\}$ is regular
- ☐ B The language $L = \{a^n \mid n \text{ is prime}\}$ is regular.
- ☐ C The language $L = \{w \mid w \text{ has } 3k + 1 \text{ b's for some } k \in \mathbb{N} \text{ with } \Sigma = \{a, b\}\}$ is regular
- ☐ D The language $L = \{ww \mid w \in \Sigma^* \text{ with } \Sigma = \{0, 1\}\}$ is regular.

#Q. To Prove a Language L is Non-Regular

1. Assume L is Regular
 2. There exist F.A for L and n is no. of states in that F.A
 3. Select some string W from L such that $|W| > n$.
 4. Divide W into XYZ such that $|xy| \leq n$ and $|y| > 0$.
 5. Find a suitable integer i such that UV^iW is not belongs to L .
- Then L is not Regular.

THANK - YOU